



Zewail City of Science, Technology, and Innovation
University of Science and Technology
Communications and Information Engineering

“Project Title”

A Graduation Project
Submitted in Partial Fulfillment of
B.Sc. Degree Requirements in
Communications and Information Engineering

Prepared By

“Student’s Name” “Student’s ID”
“Student’s Name” “Student’s ID”

Must include the full list of supervisors and indicate the affiliation of each one.

Supervised By

“Supervisor’s Name” – “Program’s Name”
“Supervisor’s Name” – “Program’s Name”

Signature

2025/2026

Dissertation main text format:

- Use Times New Roman font with a size of 12 pt.
- Headings 14 pt, and subheadings should be of the same font style and size with bold letters.
- Subheadings should start with the chapter number followed by numbers of the sub-sections separated by dots. (example: 2.1.3. Chapter 2, section 1, subsection 3:).
- Line spacing: 1.5 lines.
- No extra spacing should be present between paragraphs.
- Use no indent, or the same indent for the first line of all paragraphs (a tab or 5, 6, or 7 spaces).
- Text throughout each chapter should be continuous. Do not use frames
- Table headings and figure captions should be of the same font (Times New Roman), size 11, and single-spaced.
- No extra spacing should be present before or after figures and paragraphs.
- Do not start a table or a figure after the middle of the page unless it will be complete in the same page (whenever possible, do not split tables).
- If any table takes a whole page, do not start this page with a paragraph that will push part of this table to the next page. Instead, dedicate a whole page for the table to be complete.
- All paragraphs should be justified to the left.
- Normal margins should be used with 1 inch from top, bottom, right, and left.
- Page number should be inserted in the middle bottom of each page starting from the introduction. Roman numbers should be used for the sections before the introduction.
- Reference lists, lists of tables and figures, and contents are typed single-spaced.
- The document should be formatted to the standard US letter 8.5" x 11" page.
- Title is in all caps

Content:

- Cover page (Use the template provided on the first page).
- Acknowledgement (Acknowledge the work of those who contributed to your thesis in a direct or indirect way).
 - o In the case of a jointly supervised SDP, the full list of the SDP advisors and their affiliations (CIE and non-CIE) should be acknowledged in this section.
- Declaration
 - o You need to declare that:
 - The thesis has been composed by the students whose names and ids are shown on the title page, and
 - The thesis represents your own work (In case you did this project as part of a larger project, you need state your contribution (it should be a substantial contribution)), and
 - The work has not been submitted for any other degree or professional qualification except as specified.
- English abstract. Abstract between 200-350 words, sections include: motivation, problem, proposed solution, methods, results, and conclusions.
- Table of Contents.
- List of tables (optional).
- List of figures (optional).
- List of abbreviations (optional).
- Introduction.
- Literature Review.
- Proposed Design.
- Implementation
- Discussion and Project impact.
- Conclusion.
- References.
- Arabic abstract.
- Arabic cover page.
- Appendices (optional): Include all the details that you couldn't include in your report. Items include:
 - o List of Components
 - o Project cost sheet
 - o Support letters from industry partners (if applicable)
 - o Funding letters (if applicable)
 - o User Manual (if applicable)
 - o Software code, circuit diagrams, drawings, data sheets, and photos.

Description of the final report Sections

Your final report for CIE 599 is an 80 -120- page document) that includes the following elements:

Introduction (4-6 pages):

An introduction to your report (should be understood by non-technical readers as well as technical readers). The introduction has the following sub-elements:

- **Motivation:** why it is important to solve the problem you are targeting. The benefits you are bringing to the society or the industry. Motivation may consider one or more of the following factors
 - **Economic and global context:** What are the products/ services related to your project? What is the size of the market? Who is interested in solving this problem (target customer)? What does it mean to solve your problem economically and globally?
 - **Societal and environmental context:** Does your product have a positive impact on the society and environment.
 - **Cultural:** Will your product work in different cultures? Do you have expectations about how people respond to your innovation (include evidence from the literature if possible)?
- **Overview of the key challenges:** a broad discussion of why solving this problem is challenging (you may include some evidence from your literature review).
- **Project scope:** You need to narrow down and focus on the aspects of the problem that you are addressing in your project. State these aspects clearly highlighting the key methods you investigated to solve the problem and any related assumptions.
- **Proposed methodology:** A discussion of the proposed methodology and how you are planning to implement it. the discussion shall be presented in a way that can be understood by non-technical audience.
- **Summary of the implementation and results:** A discussion of the prototype you implemented, tools you used for this implementation and your final results.
- **Report Organization:** A summary of how your report is organized in the following chapters.

Literature review (10-13 pages):

- **Market Survey:** If you are introducing a new tool/product, you need to survey the tools and techniques that exist in the market. You need also to survey the tools/ techniques that are incorporated in your project and a justification for choosing them.
- **Existing research efforts:** A survey of the state of the art concerning the subject under consideration. You need to summarize the key research questions and theoretical paradigm. You need also to discuss the findings and conclusions of pertinent literature.
- **Best Practices and Standards:** A survey of standards and best practices related to the implementation of the different functions in your project. If no standards are available,

you should clearly state it and provide a discussion of standards under development and committees working to address the standard gaps in the area under investigation.

- **Related laws and regulations:** You need to provide a discussion of laws, codes and regulations that have a direct impact on your project
- **Related Code of Ethics:** You need to discuss a minimum related code of Ethics that you will consider in your implementation (e.g. the software engineering code of ethics <https://ethics.acm.org/code-of-ethics/software-engineering-code/>)
- **Conclusion:** this chapter should be ended with a summary of what you learn from the literature review and a description of the technical approach you intend to use in the design of the project. You may also include any gaps that you identified as part of your survey.

Proposed Design (25–40 pages):

In this deliverable, you need to discuss the methodology and the design elements. Please submit a 25–40-page document including the following elements:

- **Problem definition and project purpose:** Write a clear and concise problem statement identifying the challenge you are trying to solve in your project. The scope of the project (what you are solving and what you are not considering at the moment) should be defined clearly in this subsection.
- **Selected standards and how they are used in the design:** A detailed description of the standards you will use in your projects and how you are planning to use them.
- **Identify the major concerns in global, economic, environmental, and societal contexts that has a direct impact on your design.**
- **Requirements and technical specifications:** Starting from the problem statement and the scope of your project, provide a detailed discussion of the functional and non-functional requirements of your project.
- **Constraints:** You need also to include any time/budget constraints. Also, any constraints imposed by standards, laws, regulations and engineering codes.
- **Risks:** A discussion of the reasons why your project may fail and how you are planning to mitigate these risks.
- **Design alternatives and justification:** Typically, you have several design alternatives to implement the features you introduce in the requirements section. Discuss these alternatives comparing their weaknesses and strengths. Provide a justification of the design approach(es) you use in your implementation.
- **Analysis of the design alternatives and a justification of the design approach:** Use a decision analysis technique to select the proper design and highlight why your choice makes sense (you may for example discuss the feasibility of the solution and its advantages compared to other design choices).
- **Description of the selected design:** illustration of the final design and the approach you use to implement the different features. Your description shall include an overview of the system you are currently building, block diagram illustrating the key functions of the subsystems/ components of your design and the interaction between these subsystems/ components. You should also use suitable modeling techniques (e.g. mathematical equations, graphical representation, etc.) to represent the problem.

- *A discussion of how your proposed design adheres to recognized codes of ethics, regulations and laws that you identified in your literature review.*
- *A discussion of how your proposed design responds to the concerns you previously identified in global, economic, environmental, and societal contexts .*

Optional, you may add the following:

- **Research Questions and Hypotheses:** In case of research and development (R&D) projects, include a detailed discussion of the research questions you are trying to answer. You should also state any hypotheses you are making as a starting point.
- **Background:** any background needed about the methodology you use in your research.
- **Procedure:** Discuss the procedure for implementing your project including: data collection methods, your case study selection (if any), how do you intend to analyze your results.

Implementation (30–40 pages):

- **Solution Overview:** A discussion of the key modules of your proposed solution. Discussion may include a block diagram illustrating the key components of your implementation and the function of each module.
- **Implementation Procedure:** A detailed discussion of the implementation procedure used in each module of your system. The discussion shall include the following:
 - Demonstration of the iterative process you used to develop and improve each component of your solution.
 - Discussion of the data sources, equipment, software tools, software libraries and protocols used in your implementation. Justify your selection.
 - Schematics, flowcharts, pseudocode, material selection, manufacturing/ assembling processes, and any other information related to the implementation.
- **Testing and evaluation:** Discuss all the tests you had to validate your implementation including the unit test, integration tests and any other performance testing used in your implementation. In your discussion, you need to:
 - Discuss the test cases and scenarios you used to validate your approach
 - Demonstration how you used appropriate computer-based tools, statistics, and other resources effectively to analyze, validate, and interpret experiment results
 - Compare your results to theoretical predictions
- **Implementation timeline:** Work breakdown structure that includes all the tasks you have worked on, and the time for each task.
- **A discussion of how standards and best practices are used in your implementation.**
- **Discussion and Project impact:** An assessment of the impact of proposed solution that includes the following elements:
 - **Discussion of the performance of your implemented solution:** Comment on the results of the testing procedure you highlighted in the implementation section. Discuss the strengths and weaknesses of the proposed solution
 - **Economic impact and sustainability:** you may consider the revenues that the product may generate, costs of implementing and maintaining and any other factors you find useful. You need also to discuss sustainability. That is, after the

completion of the project, how the project will sustain itself. A detailed cost-to-benefit analysis is recommended.

- **Societal impact:** Does your project has an impact on the society. Support your arguments using examples from the literature or the daily news. Discuss how the project affect the health, safety and welfare of the society.
- **Environmental impact:** Does your project have an impact on the environment? Why and why not? You should include the impact during the implementation and when the product is released and used by your customers.
- **Cultural impact:** How does your project affect the way people live, work, socialize and organize themselves?
- **Conclusion (3-5 pages):** A conclusion of your report. You need to discuss:
 - Summary of the proposed solution and testing results.
 - Things you have learned and the limitations of your system.
 - Things you need to do in the future to address the limitations of your system.